

ANANYA SRIDHAR

Robotics & Computer Vision Engineer — Autonomous Systems, Simulation & Edge CV

Mumbai, India · +91 8104921962 · ananyasridhar108@gmail.com

github.com/darayavahoush · darayavahoush.github.io · linkedin.com/in/ananya-sridhar-ai

PROFESSIONAL SUMMARY

Robotics and Computer Vision Engineer with specialized expertise in physics-backed simulation (MuJoCo), multimodal sensor fusion (Extended Information Filter), Deep Reinforcement Learning, and real-time edge perception (OpenCV, DINOv2). Proven track record designing autonomous navigation agents, sub-50ms CPU vision pipelines, and interactive 3D robot kinematics simulators.

TECHNICAL SKILLS

- Robotics & Simulation:** MuJoCo 3D Physics, Gymnasium, Sensor Fusion (EIF/EKF), Lidar SLAM / Motion Planning, Forward & Inverse Kinematics, Three.js, React Three Fiber
- Computer Vision:** OpenCV, MediaPipe Tasks, DINOv2 (Vision Transformers), Optimal Transport (POT), Background Subtraction, Multi-Object Centroid Tracking, Face Recognition
- AI & Deep Learning:** PyTorch, Deep Reinforcement Learning (PPO, Q-Learning), MCTS, A* Pathfinding, Scikit-Learn
- Systems & Backend:** Python (Proficient), C++, C, FastAPI, Uvicorn, WebSockets, Docker, Git, Linux, ROS Concepts

PROFESSIONAL EXPERIENCE

Manas Learning — AI/ML Engineer Intern (Perception & Systems)

Mar 2026 – Present | Remote / Mumbai, India

- Real-Time Visual Biofeedback:** Integrated MediaPipe Tasks Vision for real-time oral-motor landmark tracking and articulatory movement verification during speech practice.
- Reinforcement Learning Agent:** Developed a Q-learning adaptive difficulty controller dynamically tuning game environments based on user accuracy and reaction times.
- High-Performance Architecture:** Delivered a modular FastAPI and WebSocket backend streaming low-latency perceptual feedback to React client applications.

Nivesh AI — AI/ML Engineer Intern

Jun 2026 – Present | Mumbai, India

- Streaming Audio & Telemetry Processing:** Engineered scalable data pipelines on Azure handling streaming audio feeds and document ingestion for 10,000+ mutual fund distributors.
- Spectral Noise Reduction:** Implemented digital signal preprocessing (noisereduce) and acoustic filtering pipelines to isolate voice signals in high-noise environments.

FEATURED PROJECTS

NIRVANA (Firebot) — Autonomous Firefighting Robot & MuJoCo Simulation

[GitHub: Firebot](#) · [Live Console & Docs](#)

Python, MuJoCo 3D, PyTorch, Gymnasium, EIF Sensor Fusion, Lidar SLAM, FastAPI, WebSockets

- Developed an autonomous firefighting robot operating in a physics-accurate MuJoCo 3D simulator across procedural 8-room environments with volumetric particle fire modeling.
- Designed **MM-FusionRL** (multimodal cross-attention deep RL) coupled with an **Extended Information Filter (EIF)** for thermal localization, lidar motion planning, and autonomous obstacle navigation.
- Built a web mission console streaming real-time robot telemetry, lidar sweeps, and fire-event logs via WebSockets.

Tripwire — Real-Time CPU-Only Computer Vision Vehicle Counter

[Live Demo: tripwire-inky.vercel.app](#) · [GitHub](#)

Python, OpenCV, FastAPI, React, Vite, WebSockets, Docker

- Engineered an edge-ready, GPU-free computer vision pipeline processing live CCTV and RTSP feeds with sub-50ms latency.
- Implemented Gaussian background subtraction, morphological noise filtering, centroid-based multi-object tracking, and directional virtual tripwire crossing counters streamed over WebSockets to a Vite console.

GECO-Enhanced — Confidence-Aware Semantic Keypoint Correspondence

[GitHub: GECO](#)

PyTorch, DINOv2 (ViT), Python Optimal Transport (POT), FastAPI, React, Docker

- Formulated an unsupervised semantic correspondence framework establishing keypoints across different instances of identical object classes under severe pose and perspective variation.
- Extracted dense DINOv2 ViT features and solved correspondence using Sinkhorn-regularized **Optimal Transport (POT)**.

3-DOF Robotic Manipulator Simulator — Kinematics & Optimization

[Live Demo: robotic-manipulator-bice.vercel.app](#) · [GitHub](#)

Python, FastAPI, NumPy, SciPy, Three.js, React, Vite, Recharts

- Developed an interactive 3D kinematics simulator for 3-DOF planar and spatial robotic arms (RRR and PPP configurations).
- Implemented forward and inverse kinematics solvers using SciPy numerical optimization for precision end-effector trajectory tracking; rendered reachable workspace boundaries and joint torques in Three.js.

EDUCATION & CERTIFICATIONS

- Bachelor of Technology (B.Tech)**, SASTRA University (Expected 2027)
- SWAYAM – IIT Madras:** Comprehensive Electronics and Embedded Systems (*In-Person Practical Program*)
- IIoT Certification:** Ministry of Heavy Industries (MHI) Program (*Project-Based Evaluation*)